Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A rubber ball is dropped from a height of $5\ \text{m}$ on a planet where the acceleration due to gravity is not known. On bouncing it rises to $1.8\ \text{m}$. The ball loses its velocity on bouncing by a factor of

(A) $\frac{16}{25}$

(B) $\frac{2}{5}$

(C) $\frac{3}{5}$

(D) $\frac{9}{2}$

Q.2

A ball of mass m moving with a velocity v_1 strikes normally a massive rock of mass $M(>> m)$ moving towards the ball with a velocity v_2. If the collision is perfectly elastic, the magnitude of the velocity of the ball after impact is

(A) $v_1 + v_2$

(B) $v_1 - v_2$

(C) $v_1 + 2v_2$

(D) $v_1 - 2v_2$

Q.3

Ball A moving with momentum p undergoes a one dimensinal collision with a stationary ball B of the same mass. During the collision ball B imparts an impulse I to ball A. The coefficient of restitution is

(A) $\frac{2I}{p}$

(B) $\frac{2I}{p} - 1$

(C) $\frac{I}{p} + 1$

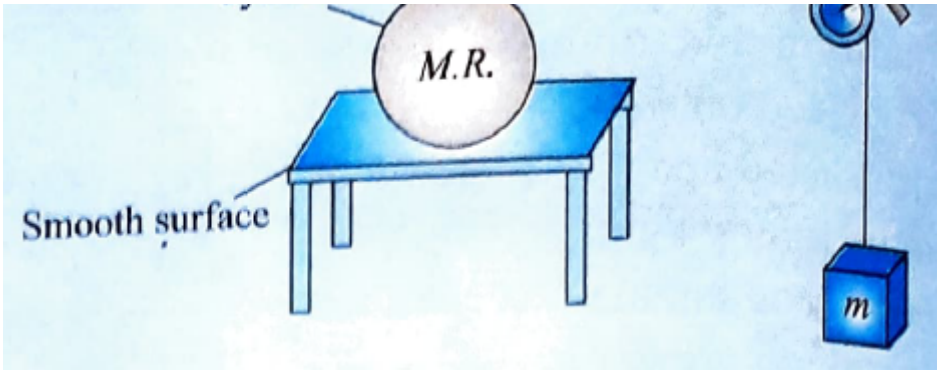
(D) $\frac{2I}{p} + 1$

Q.4

An inextensible thread is wound round a cylinder of mass $M = 7.0\ \text{kg}$ and radius $R = 10\ \text{cm}$. The cylinder is placed over a smooth horizontal surface and thread is passed over a massless and frictionless pulley such that the part of the thread between cylinder and pulley is horizontal as shown in figure.
A block of mass $m = 1.0\ \text{kg}$ is attached with free end of thread. If the system is released from rest, calculate acceleration of the block. $\left(g = 10\ \text{m s}^{-2}\right)$.



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(A) $8\ \mathrm{ms}^{-2}$

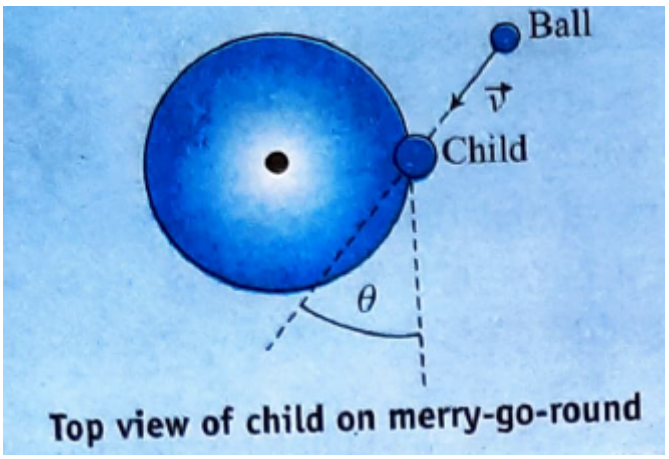
(B) $5\ \mathrm{ms}^{-2}$

(C) $6\ \mathrm{ms}^{-2}$

(D) $3\ \mathrm{ms}^{-2}$

Q.5

A child of mass $39\ \mathrm{kg}$ stands on the edge of a stationary merrygo-round of radius $2.5\ \mathrm{m}$. The moment of inertia of the merrygo-round about its rotation axis is $150\ \mathrm{kg\ m}^2$. The child catches a ball of mass $1.0\ \mathrm{kg}$ thrown by a friend. Just before the ball is caught, it has a horizontal velocity of magnitude $50\ \mathrm{m/s}$, at angle $\theta = 37^\circ$ with a line tangent to the outer edge of the merry-go-round, as shown. What is the angular speed of the merry-go-round just after the ball is caught?



(A) 0.25 rad/s

(B) 0.23 rad/s

(C) 0.29 rad/s

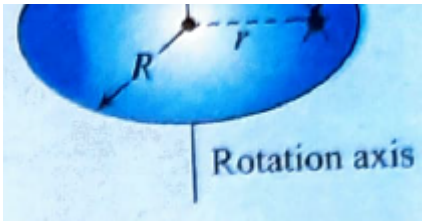
(D) 0.20 rad/s

Q.6

In figure, a cockroach with mass m rides on a disk of mass $6.00\ \mathrm{m}$ and radius R. The disk rotates like a merry-go-round around its central axis at angular speed $\omega_i = 1.50\ \mathrm{rad/s}$. The cockroach is initially at radius $r = 0.800R$, but then it crawls out to the rim of the disk. Treat the cockroach as a particle. What then is the angular speed?



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(A) 1.37 rad/s

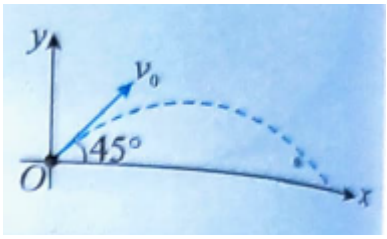
(B) 1.47 rad/s

(C) 1.34 rad/s

(D) 2.37 rad/s

Q.7

A particle is projected at time $t = 0$ from a point O with a speed v_0 at an angle of 45° to the horizontal. Find the magnitude and the direction of the angular momentum of the particle about the point O at time $t = v_0/g$.



(A) $\frac{mv_0^2}{2\sqrt{2}g}(-\hat{k})$

(B) $\frac{mv_0^2}{3\sqrt{2}g}(-\hat{k})$

(C) $\frac{mv_0^2}{4\sqrt{2}g}(-\hat{k})$

(D) $\frac{mv_0^3}{2\sqrt{2}g}(-\hat{k})$

Q.8

A particle moving in a circular path has an angular momentum of L. If the frequency of rotation is halved, then its angular momentum becomes

(A) $\frac{L}{2}$

(B) L

(C) $\frac{L}{3}$

(D) $\frac{L}{4}$

Q.9

A particle of mass $2\ \text{kg}$ is on a smooth horizontal table and moves in a circular path of radius $0.6\ \text{m}$. The height of the table from the ground is $0.8\ \text{m}$. If the angular speed of the particle is 12rads^{-1}, the magnitude of its angular momentum about a point on the ground right under the centre of the circle is

(A) $14.4\ \text{kg m}^2\ \text{s}^{-1}$

(B) $8.64\ \text{kg m}^2\ \text{s}^{-1}$

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(C) $20.16\ \text{kg m}^2\ \text{s}^{-1}$

(D) $11.52\ \text{kg m}^2\ \text{s}^{-1}$

Q.10

If the angular momentum of a particle of mass m rotating along a circular path of radius r with uniform speed is L, the centripetal force acting on the particle is

(A) $\frac{\text{L}^2}{mr^3}$

(B) $\frac{\text{L}^2}{mr}$

(C) $\frac{\text{L}}{mr}$

(D) $\frac{\text{L}^2 m}{r}$

Q.11

A particle moving in a circular path has an angular momentum of L. If the frequency of rotation is halved, then its angular momentum becomes

(A) $\frac{\text{L}}{2}$

(B) L

(C) $\frac{L}{3}$

(D) $\frac{\text{L}}{4}$

Q.12

A particle of mass $2\ \text{kg}$ is on a smooth horizontal table and moves in a circular path of radius $0.6\ \text{m}$. The height of the table from the ground is $0.8\ \text{m}$. If the angular speed of the particle is 12rads^{-1}, the magnitude of its angular momentum about a point on the ground right under the centre of the circle is

(A) $14.4\ \text{kg m}^2\ \text{s}^{-1}$

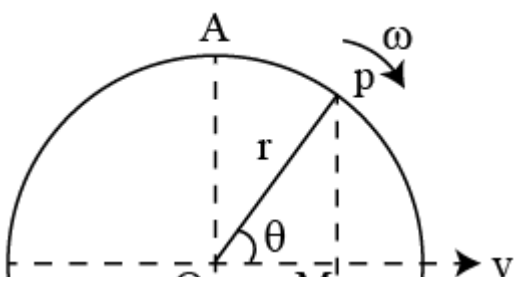
(B) $8.64\ \text{kg m}^2\ \text{s}^{-1}$

(C) $20.16\ \text{kg m}^2\ \text{s}^{-1}$

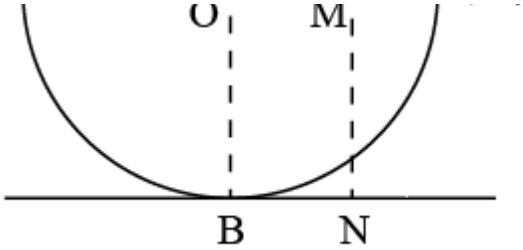
(D) $11.52\ \text{kg m}^2\ \text{s}^{-1}$

Q.13

Fig. shows a disc rolling on a horizontal plane with linear velocity v. Its linear velocity is v and angular velocity is ω. Which of the following gives the velocity of the particle P on the rim of the disc?



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(A) $v(1+\cos\theta)$

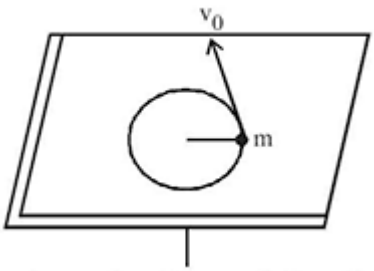
(B) $v(1-\cos\theta)$

(C) $v(1+\sin\theta)$

(D) $v(1-\sin\theta)$

Q.14

A mass m moves in a circle on a smooth horizontal plane with velocity $\mathrm{v_0}$ at a radius $\mathrm{R_0}$. The mass is attached to string which passes through a smooth hole in the plane as shown. The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{\mathrm{R_0}}{2}$. The final value of the kinetic energy is



(A) $\frac{1}{4}\mathrm{mv_0^2}$

(B) $2\mathrm{mv_0^2}$

(C) $\frac{1}{2}\mathrm{mv_0^2}$

(D) $\mathrm{mv_0^2}$

Q.15

A solid sphere, disc and solid cylinder all of the same mass and made of the same material are allowed to roll down (from rest) on an inclined
plane, then

(A) solid sphere reaches the bottom first

(B) solid sphere reaches the bottom last

(C) disc will reach the bottom first

(D) all reach the bottom at the same time

Q.16

In a bicycle, the radius of rear wheel is twice the radius of front wheel. It r_f and r_r are the radii and v_f and v_r are the speeds of topmost points of wheels then

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(A) $v_r = 2v_f$

(B) $v_f = 2v_r$

(C) $v_f = v_r$

(D) $v_f = 4v_r$

Q.17

A solid cylinder of mass $50\ \text{kg}$ and radius $0.5\ \text{m}$ is free to rotate about the horizontal axis. Amassless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 revolutions s^{-2} is

(A) $25\ \text{N}$

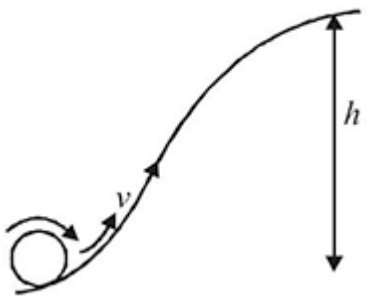
(B) $50\ \text{N}$

(C) $78.5\ \text{N}$

(D) $157\ \text{N}$

Q.18

A solid sphere is rolling on a surface as shown in figure, with a translational velocity $v\ \text{ms}^{-1}$. If it is to climb the inclined surface continuing to roll without slipping, then minimum velocity for this to happen is



(A) $\sqrt{2gh}$

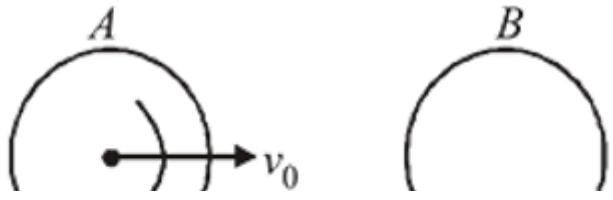
(B) $\sqrt{\frac{7}{5}gh}$

(C) $\sqrt{\frac{7}{2}gh}$

(D) $\sqrt{\frac{10}{7}gh}$

Q.19

A hollow smooth uniform sphere A of mass m rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of B to that of A just after the collision is



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(A) $1:1$

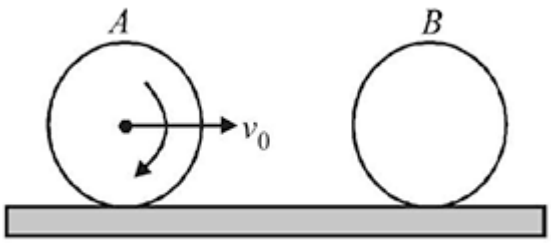
(B) $2:3$

(C) $3:2$

(D) None of these

Q.20

A hollow smooth uniform sphere A of mass m rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of B to that of A just after the collision is



(A) $1:1$

(B) $2:3$

(C) $3:2$

(D) None of these

Answers & Solutions

Q.1 Answer:

$\frac{2}{5}$

Solution:

A ball dropped from a height h_1 on reaching the planet's surface will have a velocity given by

$v_1 = \sqrt{2gh_1}$

Let v_2 be the velocity with which the ball bounces. It will attain a height h_2 given by

$v_2 = \sqrt{2gh_2}$

$\therefore \quad \frac{v_2}{v_1} =$

or $1 - \frac{v_2}{v_1} = 1 - 0.6$ or $\frac{v_1 - v_2}{v_1} = 0.4 = \frac{2}{5}$

Hence the correct choice is (b).

Q.2 Answer:

$v_1 + 2v_2$

Solution:

For a perfectly elastic one-dimensional collision, the velocity of the ball after the colision is

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$v_1 = \left(\frac{m-M}{m+M}\right) u_1 + \frac{2M(-u_2)}{m+M}$

Since $M >> m$,

$v_1 = -u_1 - 2u_2 = -(u_1 + 2u_2)$

The magnitude of v_1 **is**

$v_1 \mid = u_1 + 2u_2$

Q.3 Answer:

$\frac{2I}{p} - 1$

Solution:

Let m **be the mass of each ball. Let** u_1 **and** u_2 **be the velocities** A **and** B **before collision and** v_1 **and** v_2 **are collision. Then** $u_1 = \frac{p}{m}$ and $u_2 = 0$ (given)

Impulse = change in momentum. It is given that I **= change in momentum of** B**.**

$$= mv_2 - 0$$

$$\Rightarrow \quad v_2 = \frac{I}{m}$$

Therefore, $v_1 = \frac{p-I}{m}$

Now

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$= \frac{\left(\frac{I}{m} - \frac{p-I}{m}\right)}{\left(\frac{p}{m} - 0\right)} = \frac{2I}{p} - 1$$

So the correct choice is (b).

Q.4 Answer:

$3\ \mathrm{ms}^{-2}$

Solution:

As horizontal surface is smooth, and force is acting on the e cylinder, therefore the cylinder will slip over the surface.

Let acceleration of axis of the cylinder be a_1 **(rightward and its angular acceleration be** α **(clockwise) then downward acceleration of the block will be equal to,**

$a_2 = a_1 + R\alpha$

Taking moment of forces acting on the cylinder about 0.

$T \cdot R = I\alpha$ where $I = \frac{MR^2}{2}$

$T = \frac{1}{2}MR\alpha \quad \ldots(\mathrm{i})$

For horizontal forces on the cylinder,

$T = Ma_1$ **...(ii)**

From Eqs. (i) and (ii)

$R\alpha = 2a_1$

hence $a_2 = 3a_1$

Now considering forces acting on the block,

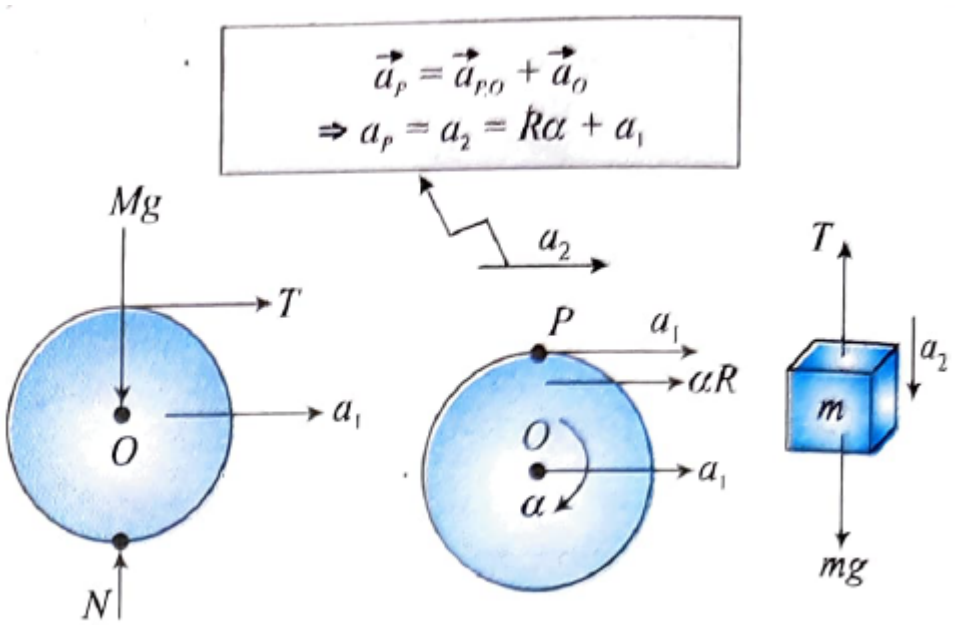
$mg - T = ma_2$

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Substituting $T = Ma_1$ **and** $a_2 = 3a_1$

$a_1 = \frac{mg}{(M+3m)}$ or $a_1 = 1\ \text{ms}^{-2}$

Hence, acceleration of the block, $a_2 = 3\ \text{ms}^{-2}$

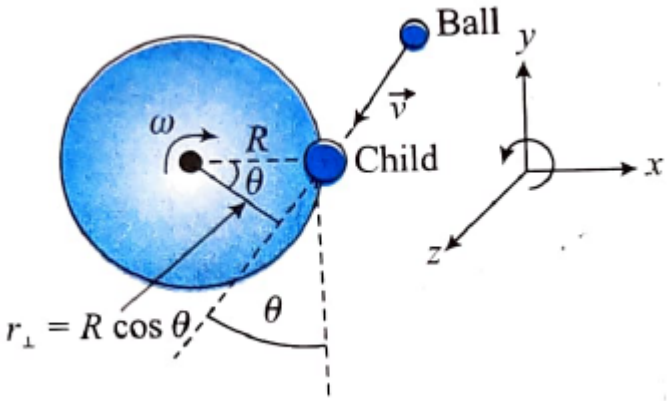


Q.5 Answer:

0.25 rad/s

Solution:

There is no net external torque acts on the system that consist of the 'child, merry-go-round and ball'. It means, the angular momentum of that system about the rotation axis must remain constant.



We define the magnitude of angular momentum of a particle,

$L = mvr_\perp$

where $r_\perp$ **is the perpendicular distance between the axis of rotation to line of action of** $\vec{p}$**.**

Let m **and** v_0 **be the mass and initial speed of the ball and** R **the radius of the merry-go-round. In this case,**

$r_\perp = R\cos 37^\circ = 2.5 \times \frac{4}{5} = 2\ \text{m}$ **...(i)**

The initial angular momentum is

$L_i = mv_0r_\perp = 1 \times 50 \times 2 = 100\ \text{kg m}^2/\text{s}$

in clockwise direction. In vector form, we can write

$\vec{L}_i = 100(-\hat{k})\text{kgm}^2/\text{s}$

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After catching the ball, the system 'child, merry-go-round and ball' rotate with common angular velocity in clockwise direction to keep angular momentum conserve. Let this angular velocity be ω, then final angular momentum

$L_f = I_{\text{system}}\,\omega$ **...(ii)**

Now let us calculate moment of inertia of the system,

$I_{\text{system}} = I_{\text{merrygoround}} + I_{\text{child}} + I_{\text{ball}}$

or, $I_{\text{system}} = 150 + (39)R^2 + (1.0)R^2 = 150 + 40R^2 = 400\ \text{kg m}^2$

Substituting the value of I_{system} **in Eq. (i), we get**

$L_f = 400\omega$

In vector form we can write,

$\vec{L}_f = 400\omega(-\hat{k})\text{kgm}^2/\text{s}$ **...(iii)**

As angular momentum is conserved then, $\vec{L}_i = \vec{L}_f$ **or,** $100(-\hat{k}) = 400\omega(-\hat{k})$**, hence** $\omega = 0.25\text{rad/s}$.

Q.6 Answer:

1.37 rad/s

Solution:

The cockroach's crawl changes the mass distribution (and thus the rotational inertia) of the cockroach-disk system. The angular momentum of the system does not change because there is no external torque to change it. (The forces and torques due to the cockroach's crawl are internal to the system.)

We want to find the final angular speed. Our key is to equate the final angular momentum L_f **to the initial angular momentum** L_i**, because both involve angular speed. They also involve rotational inertia** I**. So, let's start by finding the rotational inertia of the system of cockroach and disk before and after the crawl. The rotational inertia of a disk rotating about its central axis is given by** $I_d = \frac{1}{2}MR^2$**.**

Substituting $6.00\ \text{m}$ **for the mass** M**, our disk here has rotational inertia,**

$I_d = 3.00mR^2$ **...(i)**

We know that the rotational inertia of the cockroach (a particle) is equal to mr^2**. Substituting the cockroach's initial radius** $(r = 0.800R)$ **and final radius** $(r = R)$**, we find that its initial rotational inertia about the rotation axis is,**

$I_{ci} = 0.64mR^2$ **...(ii)**

and its final rotational inertia about the rotation axis is

$I_{cf} = mR^2$ **...(iii)**

So, the cockroach-disk system initially has the rotational inertia.

$I_i = I_d + I_{ci} = 3.64mR^2$ **...(iv)**

and finally has the rotational inertia,

$I_f = I_d + I_{cf} = 4.00mR^2$ **...(v)**

Next, we use equation $(L = I\omega)$ **to write the fact that the system's final angular momentum** L_f **is equal to the system's initial angular momentum** L_i **:**

$I_f\omega_f = I_i\omega_i$ or $4.00mR^2\omega_f = 3.64mR^2(1.50\text{rads})$

After canceling the unknowns m **and** R**, we come to,**

$\omega_f = 1.37\text{rad/s}$

Note that ω **decreased because part of the mass moved outward, thus increasing that system's rotational**

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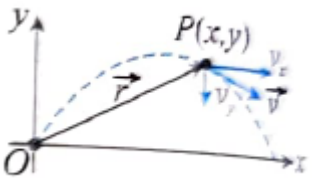
inertia.

Q.7 Answer:

$\frac{mv_0^2}{2\sqrt{2}g}(-\hat{\mathrm{k}})$

Solution:

In this case the angular momentum of the particle will not remain constant as there is always torque of gravity is acting. Let us take the origin at O, x-axis along the horizontal and y-axis along the vertically upward direction as shown in figure. For horizontal motion during the time 0 to t,



$v_x = v_0 \cos 45^\circ = \frac{v_0}{\sqrt{2}}$

and $x = v_x t = \frac{v_0}{\sqrt{2}} \frac{v_0}{g} = \frac{v_0^2}{\sqrt{2}g}$

For vertical motion, using $v_y = v_{oy} - gt$**, we get**

$$\mathrm{v}_y = v_0 \sin 45^\circ - g\left(\frac{v_0}{g}\right) = \frac{(1-\sqrt{2})}{\sqrt{2}} v_0$$

$$\text{and } y = (v_0 \sin 45^\circ)\, t - \frac{1}{2} g t^2$$

$$\Rightarrow \quad y = \frac{v_0}{\sqrt{2}}\left(\frac{v_0}{g}\right) - \frac{1}{2} g \left(\frac{v_0}{g}\right)^2 = \frac{v_0^2}{2g}(\sqrt{2}-1)$$

The angular momentum of the particle at time t about the origin is

$$L = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$

$$= (x\hat{i} + y\hat{j}) \times \left(v_x \hat{i} + v_y \hat{j}\right) = m\left(x v_y \hat{k} - y_x \hat{k}\right)$$

$$= m\left[\left(\frac{v_0^2}{\sqrt{2}g}\right) \frac{v_0}{\sqrt{2}}(1-\sqrt{2}) - \frac{v_0^2}{2g}(\sqrt{2}-1)\frac{v_0}{\sqrt{2}}\right](\hat{k})$$

$$= \frac{mv_0^2}{2\sqrt{2}g}(-\hat{k})$$

Thus, the angular momentum of the particle is $mv_0^2/(2\sqrt{2}g)$ in the negative z-direction, i.e., perpendicular to the plane of motion, going into the plane.

Q.8 Answer:

$\frac{\mathrm{L}}{2}$

Solution:

Angular momentum of particle is given by:

$\mathrm{L} = \mathrm{mr}^2\omega = 2\pi\mathrm{mr}^2\mathrm{f} \quad [\because \mathrm{W} = 2\pi\mathrm{f}]$

If frequency is halved then, $\mathrm{L}' = \mathrm{mr}^2\frac{\omega}{2} = \pi\mathrm{mr}^2\mathrm{f} \therefore \mathrm{L}' = \frac{\mathrm{L}}{2}$

Q.9 Answer:

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$14.4\ \mathrm{kg\ m^2\ s^{-1}}$

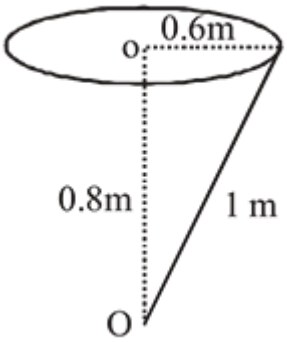
Solution:

Angular momentum, $L_0 = \mathrm{mvr}\sin 90^\circ$

$= 2 \times 0.6 \times 12 \times 1 \times 1$

[**As** $V = r\omega, \mathrm{Sin}\,90^\circ = 1]$ $\quad 0.8\ \mathrm{m}$

So, $L_0 = 14.4\mathrm{kgm^2/s}$



Q.10 Answer:

$\frac{\mathrm{L}^2}{mr^3}$

Solution:

From the relation, angular momentum, $L = mvr \quad v = \frac{L}{mr}$

Centripetal force acting on the particle $F = \frac{mv^2}{r} = \frac{m\left(\frac{L}{mr}\right)^2}{r} = \frac{L^2}{mr^3}$

Q.11 Answer:

$\frac{\mathrm{L}}{2}$

Solution:

Angular momentum of particle is given by:

$\mathrm{L} = \mathrm{mr^2\omega} = 2\pi\mathrm{mr^2f} \quad [\because \mathrm{W} = 2\pi\mathrm{f}]$

If frequency is halved then, $\mathrm{L'} = \mathrm{mr^2}\frac{\omega}{2} = \pi\mathrm{mr^2f} \therefore \mathrm{L'} = \frac{\mathrm{L}}{2}$

Q.12 Answer:

$14.4\ \mathrm{kg\ m^2\ s^{-1}}$

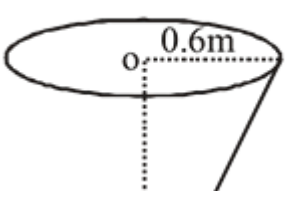
Solution:

Angular momentum, $L_0 = \mathrm{mvr}\sin 90^\circ$

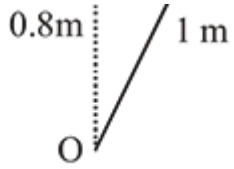
$= 2 \times 0.6 \times 12 \times 1 \times 1$

[**As** $V = r\omega, \mathrm{Sin}\,90^\circ = 1]$ $\quad 0.8\ \mathrm{m}$

So, $L_0 = 14.4\mathrm{kgm^2/s}$



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Q.13 Answer:

$v(1+\sin\theta)$

Solution:

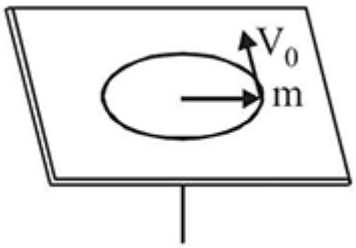
Velocity of $\mathrm{P} = (\mathrm{NP})\omega = (\mathrm{NM}+\mathrm{MP})\omega$
$= \mathrm{r}(\mathrm{r}+\sin\theta)\omega = \mathrm{v}(1+\sin\theta)$

Q.14 Answer:

$2\mathrm{mv}_0^2$

Solution:

Applying angular momentum conservation



$\mathrm{mV_0R_0} = (\mathrm{m})\left(\mathrm{V}^1\right)\left(\frac{\mathrm{R_0}}{2}\right)$
$\therefore \quad \mathrm{v}^1 = 2\ \mathrm{V_0}$
Therefore, new $\mathrm{KE} = \frac{1}{2}\ \mathrm{m}\ (2\ \mathrm{V_0})^2 = 2\mathrm{mv}_0^2$

Q.15 Answer:

solid sphere reaches the bottom first

Solution:

No solution available

Q.16 Answer:

$v_f = v_r$

Solution:

The velocity of top point of the wheel is twice that of centre of mass. And the speed of centre of mass is same for both the wheels. $[\mathrm{v} = \omega\mathrm{r}]$

Q.17 Answer:

$157\ \mathrm{N}$

Solution:

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Here $\alpha = 2$ revolutions $/s^2 = 4\pi rad/s^2$ (given)

$$I_{cylinder} = \frac{1}{2}MR^2 = \frac{1}{2}(50)(0.5)^2$$
$$= \frac{25}{4}Kg - m^2$$

As $\tau = I\alpha$ so $TR = I\alpha$

$\Rightarrow T = \frac{I\alpha}{R} = \frac{\left(\frac{25}{4}\right)(4\pi)}{(0.5)}N = 50\pi N = 157\ N$

Q.18 Answer:

$\sqrt{\frac{10}{7}gh}$

Solution:

Minimum velocity for a body rolling without slipping

$$v = \sqrt{\frac{2gh}{1+\frac{K^2}{R^2}}}$$

For solid sphere, $\frac{K^2}{R^2} = \frac{2}{5}$

$$\therefore \quad v = \sqrt{\frac{2gh}{1+\frac{K^2}{R^2}}} = \sqrt{\frac{10}{7}gh}$$

Q.19 Answer:

$3 : 2$

Solution:

After collision velocity of COM **of** A **becomes zero and that of** B **becomes equal to initial velocity of** COM **of** A**. But angular velocity of** A **remains unchanged as the two spheres are smooth.**

Q.20 Answer:

$3 : 2$

Solution:

when two identical bodies collide \& one body is at rest initially then linear momentum is totally transferred to other body.Let the velocity of sphere A **initially was** V

final velocity of $A = o$

final velocity of other sphere B **is** V

kinetic energy of $B = mV^2/2$ **.....(1)**

kinetic energy of $A = IW^2/2$

$= mV^2/3$ **.....(2) (Isphere** $= 2/3mr^2$**)**

$eq1/eq2 = 3/2$

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